

Understanding and Evaluating Routing Protocols in Computer Networks

Course: Computer Networks

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Objective

To understand the principles of routing, distinguish between different types of routing protocols, and analyze their performance and use in real-world network environments.

1. Introduction to Routing and Routing Protocols

- Routing in computer networks is the process of selecting a path for traffic in a network, or between or across multiple networks. It is a key function of the network layer (Layer 3) in the OSI model.
- Routing protocols are used to facilitate the exchange of routing information between routers, enabling them to build and maintain routing tables.
- Static routing is manually configured and remains constant unless changed by an administrator. Dynamic routing uses algorithms and routing protocols to adjust paths automatically based on network conditions.

2. Classification of Routing Protocols

Routing protocols can be classified into the following categories:

- Distance Vector Protocols: These determine the best path based on distance metrics such as hop count. Example: Routing Information Protocol (RIP).
- Link State Protocols: These use a complete map of the network to determine the shortest path. Example: Open Shortest Path First (OSPF).
- Hybrid Protocols: These combine aspects of both distance vector and link state protocols. Example: Enhanced Interior Gateway Routing Protocol (EIGRP).

3. Comparative Analysis

The following table compares RIP, OSPF, and EIGRP based on several key parameters.

Protocol	Routing Type	Convergence Speed	Metric Used	Scalability	Protocol Overhead	Use Cases & Environments
RIP	Distance Vector	Slow	Hop Count	Low	Low	Small networks
OSPF	Link State	Fast	Cost (Bandwidth)	High	High	Large enterprise networks
EIGRP	Hybrid	Faster than RIP, close to OSPF	Composite metric (Bandwidth, Delay, etc.)	High	Medium	Medium to large networks